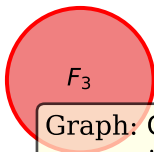
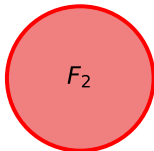
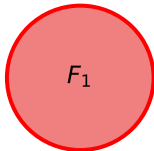


Closed-valued Multifunction

$F(\omega)$ is closed for each $\omega \in \Omega$



Graph: $\text{Gr}(F) = \{(\omega, x) : x \in F(\omega)\}$
is closed in $\Omega \times X$

Measurability Types

Strong Measurability

F is measurable (preimage of open sets)

Weak Measurability

Selections are measurable

$\mathcal{A}$ -measurability

Closed graph and measurable projections